

Modeling interception movement distributions

Seman Libbiss and Paz Flashner
Workshop on Deep Learning, Tel Aviv University
Jason Friedman and Moni Shahar

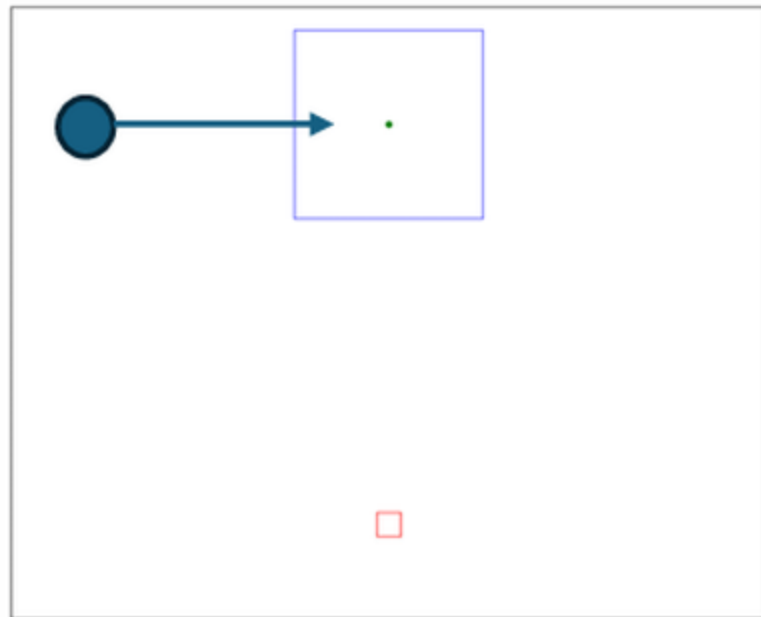
A compact personal representation for stochastic movement generation

The task and the evaluation split

4,732 trials from 28 participants

Free eye movement; target approaches from either side

Four rotations: 17 train, 4 validation, 7 test participants



Each participant enters the test set once. Median retained trials: 179.5.

The compression pipelines

Every trajectory becomes 100 x 2 positions on normalized time.

Stage	Spline + PCA	Neural models
Input	100 x 2 positions	200 standardized coordinates
Encoder	9 coefficients per axis; PCA on N x 18 matrix	Two 256-unit layers; n-value code
Decoder	Inverse PCA; two spline basis products	Two 256-unit layers; 200 coordinates
Timing output	Ridge on code + 5 conditions	Separate two-duration head

CVAE and CAE use five task conditions. VAE masks them to zero. Matched $n = 3$ and 8.

Reconstruction favors spline + PCA

The encoder receives the observed test trajectory.

Model	n = 3 MSE	n = 8 MSE
Spline + PCA	0.097255	0.022772
CAE	0.169661	0.025968
CVAE	0.176981	0.032481
VAE	0.148299	0.033487

Lower is better. Spline beats each neural model under BH-140. Spline vs CAE n=8 does not survive Holm. Neural leaders: VAE n=3 and CAE n=8.

Generation uses separate context trials

Context trials establish a personal latent center.

120 latent draws use that center and a shared training covariance.

The decoder produces new paths and durations.

$$\boldsymbol{\theta}_s = \frac{1}{|C_s|} \sum_{i \in C_s} \boldsymbol{\mu}_i, \quad \mathbf{z}^{(b)} \sim \mathcal{N}(\boldsymbol{\theta}_s, \Sigma_{\text{train}}).$$

Query trajectories and query timing labels are withheld. Task-condition metadata supplies a matched mixture where the model uses it.

Generated kinematic-feature distributions

Eleven features cover timing, speed, geometry, endpoints and speed peaks.

Model	n	Mean KS	Energy	MMD ²
Spline + PCA	3	0.3251	0.9774	0.1815
CAE	3	0.2876	0.8015	0.1414
CVAE	3	0.2778	0.7454	0.1309
VAE	3	0.2846	0.7476	0.1303
Spline + PCA	8	0.2548	0.7347	0.1153
CAE	8	0.2587	0.6202	0.1111
CVAE	8	0.2194	0.4305	0.0675
VAE	8	0.2151	0.4110	0.0639

Lower is better. VAE and CVAE beat spline on all three metrics at both dimensions under BH and Holm (140 tests). VAE n=8 has the lowest means, without a uniform significant advantage over CVAE.

Full-trajectory generation supports VAE and CVAE

The new test preserves all 100 ordered positions.

Model	n	Path energy	Path MMD ²
Spline + PCA	3	0.1068	0.0777
CAE	3	0.1272	0.0818
CVAE	3	0.1024	0.0578
VAE	3	0.1005	0.0572
Spline + PCA	8	0.0981	0.0705
CAE	8	0.0947	0.0652
CVAE	8	0.0549	0.0244
VAE	8	0.0548	0.0238

At n=8, VAE/CVAE beat spline on both metrics under BH and Holm-80, also with balanced axes. VAE vs CVAE: q=.672 (energy), 1.000 (MMD). At n=3, evidence depends on metric and scaling.

The participant's own center improves generation

We change the center while holding decoder, noise and task draws fixed.

Model	Own center	Population center	Other participants
CVAE n=3	0.2778	0.3110	0.3380
CVAE n=8	0.2194	0.2896	0.3172
VAE n=8	0.2151	0.2841	0.3144

Mean feature KS; lower is better. All VAE control comparisons: BH $q < 1.6 \times 10^{-6}$ (6 tests). All CVAE comparisons: $q < .000382$ (12 tests).

What the compact model achieves

VAE n=8 is the dashboard default; CVAE remains a comparable alternative.

Spline + PCA retains the strongest reconstruction.

Stored context resampling achieves lower mean path discrepancy than compact models.

Raw path reference	Energy	MMD ²
VAE n=8	0.0548	0.0238
Resample stored context	0.0328	0.0054
Repeat context mean	0.4621	0.2452

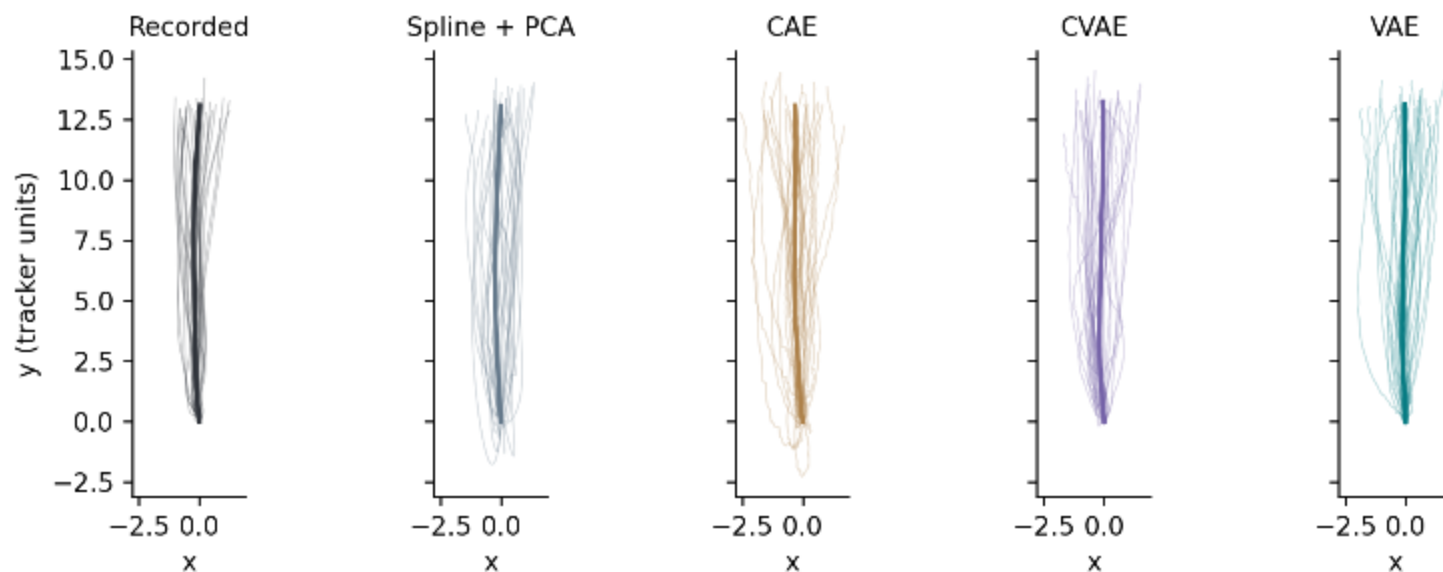
Reference differences are descriptive. Resampling retains whole paths; the mean-only reference collapses variation. Latent summary probes also leave information unrecovered.

Dashboard demonstration

Default: VAE, $n = 8$

Compare $n = 3$, CVAE and spline + PCA

Return to the benchmark view for cohort results



This backup example is subject01, $n=8$, fold 2, seed 42. Live dashboard references use fold 0, seed 42. Equal x/y scale.

Backup: resampling and the two time windows

Model input: target motion to final tracker sample.

Feature input: detected finger onset to end.

A phase of 0.5 means half the selected time interval, not half the path length.

$$\mathbf{P}_j = \mathbf{S}(j/99) - \mathbf{S}(0), \quad j = 0, \dots, 99.$$

100 samples are a common representation, not 100 independent observations.

Backup: what becomes the PCA matrix?

Object	Dimensions	One row / one column means
Sampled trial	100 x 2	Time point / x or y
Spline basis B	100 x 9	Time point / basis function
Trial coefficient vector	18	Nine x coefficients followed by nine y
Training coefficient matrix	N training x 18	Trial / coefficient
PCA directions V	18 x n	Coefficient / retained component
Training latent codes	N training x n	Trial / latent coordinate

Five interior knots + degree 3 + 1 = nine coefficients per axis.

Backup: fitting and decoding spline + PCA

Fit the two coordinate coefficient vectors by least squares.

Validation selects raw or standardized coefficients.

Inverse PCA gives coefficients; multiplying by B restores the path.

$$\mathbf{z} = V^T[(\mathbf{c} - \mathbf{m}) \oslash \mathbf{d}], \quad \hat{\mathbf{c}} = \mathbf{m} + \mathbf{d} \odot (V\mathbf{z}).$$

m and d combine centering/scaling; V contains retained PCA directions. Raw PCA uses d = 1.

Backup: predicting physical timing

Stage	Spline + PCA	Neural models
Predictors	n PCA scores + 5 conditions	Latent code + conditions where enabled
Targets	Two standardized $\log(t + .001)$ values	Same two standardized targets
Predictor	Ridge, alpha selected on validation	Separate decoder output head
Inverse	Undo scaling, exponentiate, subtract .001	Same inverse transform

Observed timing never enters the trajectory encoder. Reconstruction-time timing prediction sees the observed shape.

Backup: the neural training objective

Coordinate error sums 200 squared differences.

Timing error sums two squared differences and receives weight 20.

KL sums across latent dimensions; the batch averages the total.

$$\mathcal{L} = \|\mathbf{v} - \hat{\mathbf{v}}\|_2^2 + 20\|\mathbf{y} - \hat{\mathbf{y}}\|_2^2 + \beta D_{\text{KL}}[q(\mathbf{z} | \mathbf{v}, \mathbf{c}) \| \mathcal{N}(\mathbf{0}, I)].$$

Replacing both summed terms by separate MSE means would change the effective timing weight.

Backup: VAE, CVAE and CAE

Property	VAE	CVAE	CAE
Trajectory encoder	Two hidden layers	Two hidden layers	Two hidden layers
Task inputs	Masked to zero	Encoder and decoder	Encoder and decoder
Latent training code	Gaussian sample	Gaussian sample	Deterministic
KL penalty	Yes	Yes	No
Reconstruction code	Posterior mean	Posterior mean	Deterministic code

A nonsignificant conditional-model advantage does not mean the experimental conditions are irrelevant.

Backup: where the generated variation comes from

The personal center averages context codes.

Training residuals estimate within-person latent variation.

Variational models also add average posterior variance.

$$\Sigma_{\text{train}} = \text{Cov}_{\text{train}}(\mu_i - \bar{\mu}_{s(i)}) + \text{diag}(\mathbb{E}_{\text{train}}[\sigma_i^2]) + 10^{-6}I.$$

One shared covariance for every participant; 120 draws around each personal center.

Backup: leakage and held-out evaluation

Information	Training	Validation	Test context	Test query
Fit network/PCA/scalers	Yes	No	No	No
Select checkpoint/settings	No	Yes	No	No
Personal center	No	No	Yes	No
Reconstruction input	No	No	No	Observed path
Generation task mixture	No	No	No	Metadata only
Generation scoring	No	No	No	Paths and timings

Population model fitting and participant enrollment are separate stages.

Backup: what the eleven features measure

Group	Features
Timing	Initiation time; movement time; relative time to peak speed
Speed	Peak speed; heuristic speed-peak count
Geometry	Path length; straight-line distance; curvature ratio; maximum lateral deviation
Endpoint	Final x; final y

Generated feature computation uses predicted timing and an onset-based movement crop.

Backup: KS and joint distribution distances

KS: largest gap between two empirical cumulative curves.

Mean KS: average of eleven feature statistics.

Energy and MMD: compare whole vectors and account for joint structure.

$$D_{sf} = \sup_x |F_{sf}(x) - G_{sf}(x)|, \quad \overline{D}_s = \frac{1}{11} \sum_{f=1}^{11} D_{sf}.$$

KS statistics are not p-values. Correlated features can affect joint geometry.

Backup: energy discrepancy and mean collapse

Energy compares cross-set distances with within-set distances.
A generator needs plausible variation as well as a good center.

$$E = 2 \overline{\|X - Y\|} - \overline{\|X - X'\|} - \overline{\|Y - Y'\|}.$$

Example reference	Mean-path RMSE	Path energy
VAE n=8	0.1604	0.0548
Repeated context mean	0.1146	0.4621

Lower is better. The mean-only output has a closer mean but a much worse distribution.

Backup: Gaussian-kernel MMD

Kernel: $\exp(-\text{gamma} \times \text{squared Euclidean distance})$.

Gamma comes from the median positive training pair distance.

Within-set diagonals are excluded in the unbiased estimator.

$$\widehat{\text{MMD}}_u^2 = \frac{\sum_{i \neq j} k(X_i, X_j)}{m(m-1)} + \frac{\sum_{i \neq j} k(Y_i, Y_j)}{r(r-1)} - \frac{2 \sum_{i,j} k(X_i, Y_j)}{mr}.$$

Finite-sample unbiased MMD squared may be slightly negative. It is not an accuracy percentage.

Backup: 28 paired scores, not thousands of independent trials

First average neural seeds within each participant.

Then compare paired participant scores using two-sided Pratt Wilcoxon.

Round differences to 12 decimals before ranking.

Correction family	Tests
Original seven endpoints $\times$ ten pairs $\times$ two dimensions	140
New two path endpoints $\times$ ten pairs $\times$ two dimensions $\times$ two geometries	80
VAE replacement controls	6
CVAE replacement controls	12

BH controls false discoveries under its assumptions. Holm is a familywise-error sensitivity. Nonsignificance is not equivalence.

Backup: the boundaries of the conclusion

Useful same-session personal centers do not establish stable traits.
Direct context summaries beat latent linear probes on all 14 tested targets.
Condition routes work, but their value depends on the task and model.
Minimum-jerk decomposition remains a separate investigation.

The 31-page laboratory report retains the extended results and Jason's open questions.

Backup: launch and source navigation

Repository terminal:

```
python -m streamlit run src/confirmatory_dashboard.py --server.port 8510
```

Browser: <http://127.0.0.1:8510>

Stop: Ctrl+C in that terminal

The standalone dashboard archive contains launch instructions for use without the full research checkout.